Reg. No. [ ] [18AU54]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

FUELS, LUBRICANTS AND ADDITIVES

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	What is refinery? Explain petroleum refining process with neat sketch.	**10**
	b.	Write a short note on polymerization process.	**04**
2.	**a.**	List the requirements, types and classification of lubricating oils.	**08**
	b.	What is the importance of Viscosity and Viscosity Index in selection of Grease? Explain its significance.	**06**
3.	**a.**	Discuss the effect of engine variables on mechanical friction.	**08**
	b.	Explain the functions of lubricating system.	**06**
		OR	
4.	**a.**	Explain lubrication regimes with neat sketch.	**08**
	b.	Differentiate between hydrostatic and hydrodynamic lubrication with relevant sketch.	**06**
5.	**a.**	When 0.5g of a fuel on complete combustion in excess oxygen, increases temperature of water in a calorimeter containing 1000g of water to 2.10^{0}C. Calculate gross (higher) calorific value and net calorific value of the fuel if the fuel contains 7% of H2 and the water equivalent of the calorimeter is 120g.	**08**
	b.	Discuss the factors influencing the rate of combustion.	**06**
		OR	
6.	**a.**	Explain the following terms i. Aniline Point ii. Flammability iii. Vapour pressure iv. API Gravity	**08**
	b.	Fuel oil has the following analysis C=90%; H=6% S=2.5%, O=1%, ash=0.5%. Calculate the quantity of air required for complete combustion of 1 kg of the fuel if 30% excess air is used.	**06**
7.	**a.**	Explain the various stages of combustion in SI engine with neat sketch.	**10**
	b.	Describe the phenomenon of knocking in SI Engine.	**04**
		OR	
8.	**a.**	Discuss the effect of engine operating variables on the engine knocking.	**10**
	b.	Explain the following terms i. Ignition Lag ii. Flame propagation	**04**